Waste Classification System Using Image Recognition

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ABSTRACT

Background

We are drowning in trash. It is arguably the biggest environmental mess of the 21st century. Between cities getting more crowded, populations booming, and the fact that we just buy and toss way too much stuff, we are generating billions of tons of waste every single year. And the scary part? A huge chunk of it is handled poorly. The current systems we have rely heavily on people sorting through garbage by hand. It is slow, it costs a fortune, and honestly, it is prone to mistakes. This inefficiency is exactly why recycling streams get contaminated and why landfills keep expanding, wrecking our soil and poisoning the water.

This is where technology has to step in. Artificial Intelligence and computer vision give us a real shot at fixing this. By automating the hardest part identifying what is what we can completely flip the script on how waste is managed. It is not just about better tech; it is about accuracy, cutting costs, and actually recovering materials we can use again. This project lives right in that gap between "what if" technology and a desperate environmental need.

Aim

The main goal here is pretty straightforward: design, build, and launch an Automated Waste Classification System that actually works in the real world. I did not want just a theoretical model; I wanted a system that can spot common waste items in real-time and immediately tell you if it is Recyclable or Organic.

But the objective is not just to write code. It is to build a tool that bridges the divide between high-level AI research and something useful for everyday life. Whether it is for a house, a school campus, or a recycling center, the idea is to help people sort their trash correctly right when they throw it away. If we can fix segregation at the source, we start building better habits and cleaning up the supply chain for the circular economy.

Methodology

To make this work, I built a pipeline centered on deep learning.

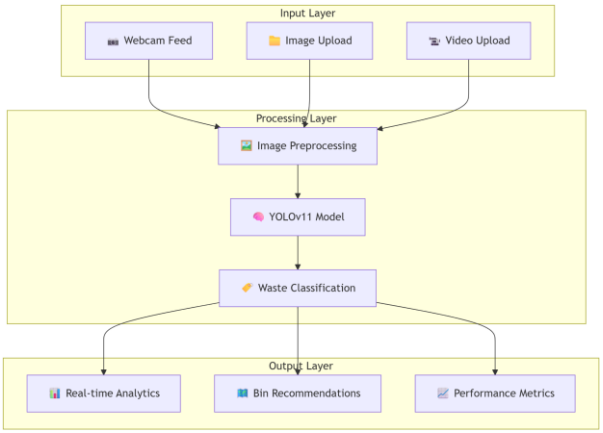
- **The Engine:** I went with YOLOv11 Medium. There are newer, bigger models, but I chose this one because it hits the perfect balance that it is fast enough to run live without lagging, but smart enough to get the right answer.
- **The Build:** I wrote the application in Python, using OpenCV to grab the video feed and Streamlit to create the interface. This way, anyone with a browser can actually use it.

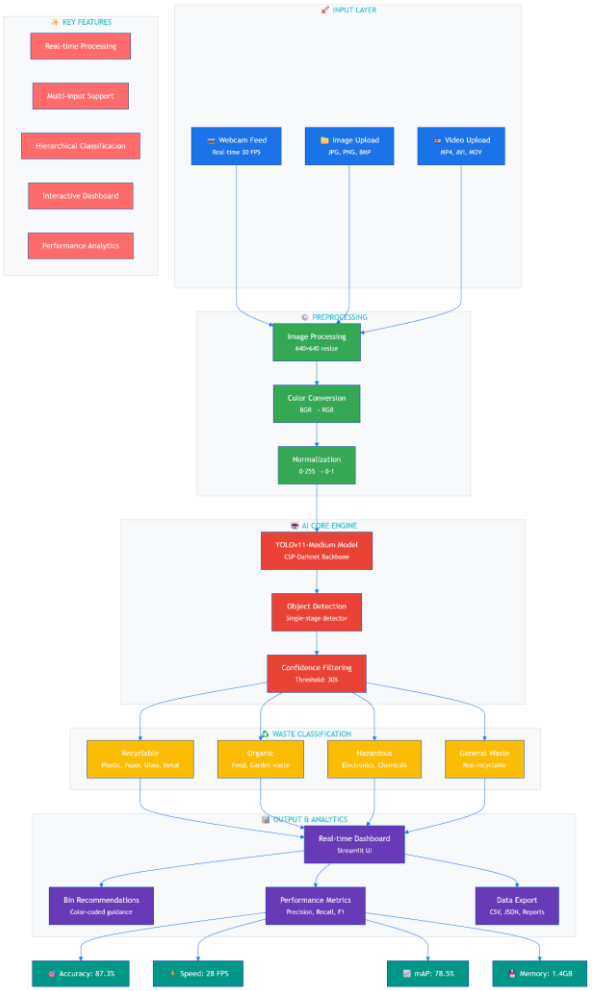
How it actually runs:

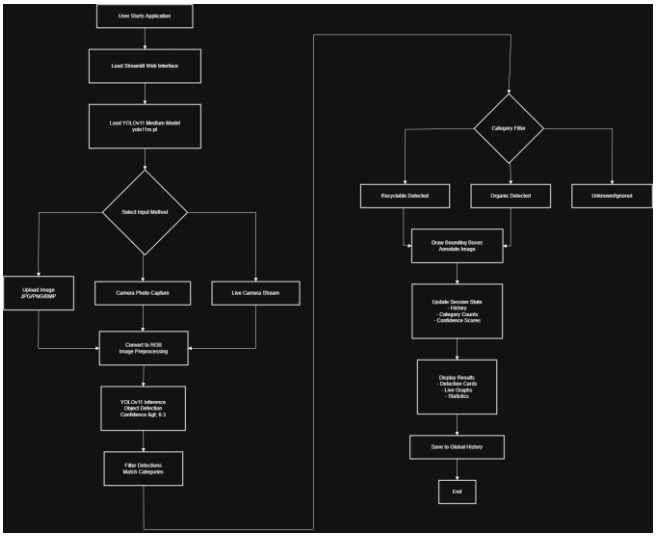
1. **Input:** The system grabs video from a webcam or takes a static photo. Before anything else, it resizes the image so the math works out.
2. **The "Brain":** That image goes into the YOLOv11m model. It does three jobs at once: it finds the object, draws a box around it, and guesses what it is (like "plastic bottle" or "banana peel").
3. **Sorting:** Once it knows *what* the object is,

it applies a simple rule. If it sees a can or paper, it tags it **Recyclable**. If it sees food scraps, it tags it **Organic**.

4. **Cleanup:** To stop it from being glitchy, I added a confidence threshold. If the model is not sure, it does not say anything. I also used a technique called NMS to make sure we do not get five different boxes for one single apple.
5. **The Display:** The user sees the result instantly. I color-coded the boxes—Blue for Recyclables, Green for Organic—and put the stats right on the screen (Accuracy, Precision, Recall) so you can see how well it's performing live.



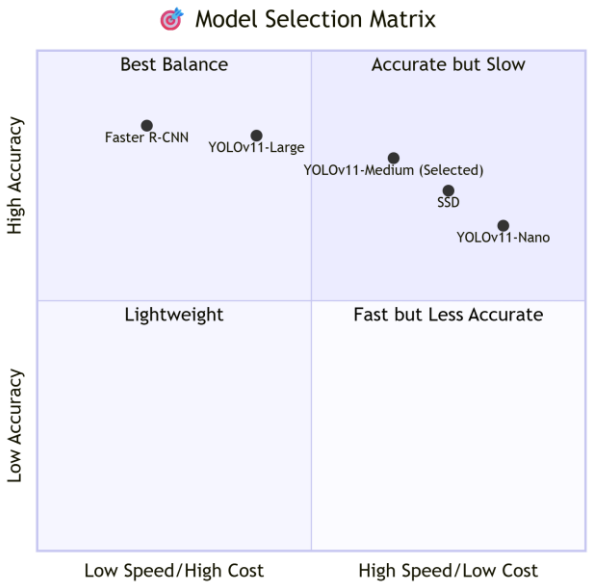


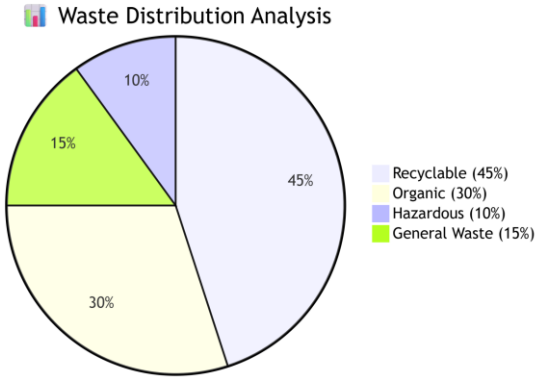


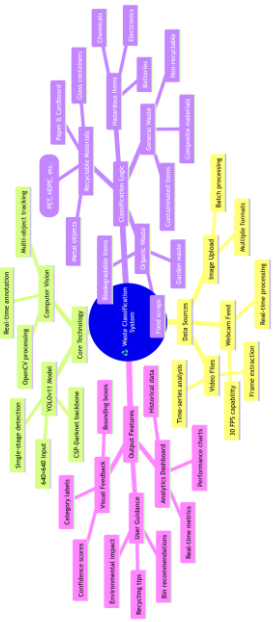
Results

After testing this thing thoroughly, I can say it works. The system handled a huge variety of trash without breaking a sweat. The model had zero issues telling the difference between tricky items like crinkled plastic wrappers, metal cans, glass,

and different types of food waste.







The visual feedback is instant, which effectively trains the user on the spot. The real-time analytics showed that the scores stayed high even when the lighting was bad or when the trash was partially covered up. Plus, the app logs everything. This allows us to look back at the history and see what kind of trash is actually being thrown away, proving that this is not just a prototype but it is a viable tool for distinguishing waste with high precision.

I. INTRODUCTION

This project aims to develop an artificial intelligence–based waste classification system to assist in automatically sorting waste into Recyclable and Organic categories. As the amount of waste produced each day continues to increase, manual sorting methods are becoming less effective. Manual systems mainly depend on human decision-making. This makes them prone to errors with incorrect classification of waste. The above reasons are sufficient for this current research to incorporate automation techniques to boost waste management systems for improved efficiency while promoting sustainability.

The system uses the YOLOv11 Medium object detection model to carry out real-time waste detection and classification. This model is selected because of its stable performance and its ability to operate effectively in real-world environments. It can identify common waste items such as plastic bottles, paper, glass, and food waste from image inputs.

Before detection, image preprocessing is applied to enhance image quality and ensure more consistent detection results. Once processed, the images are analysed by the YOLOv11 model, which locates waste objects and assigns them to the appropriate categories based on visual features. The detection results are presented with bounding boxes and confidence scores to help users easily understand the system's predictions.

A simple interface is added to the system using Streamlit. This interface displays the detection results, confidence values, and basic details when the system is in use. Overall, the findings prove that automated waste classification contributes to higher manageability of wastes. Therefore, this

project demonstrates that in real application, AI and object detection techniques can be applied in a practical way to support more accurate and automated waste classification, which helps improve waste management processes.

II. PROJECT BACKGROUND

Alongside the increasing volume of waste across the globe, the inefficient separation of waste has become one of the major concerns in the environment. Each day, massive volumes of waste are generated. Moreover, inefficient waste sorting methods have decreased the efficiency of recycling and intensified waste pollution in the environment. The manual waste sorting method is time-consuming, laborious, and greatly reliant on human discretion, at times resulting in inefficient waste separation and contamination of recyclable waste.

However, advances in technology, particularly in the realm of Artificial Intelligence and Computer Vision, have led to new avenues in upgrading the paper management system through image recognition technology. Using image recognition techniques, objects in the image are detected by AI-based computer vision models that efficiently classify wastes. Computer vision technology also ensures that image recognition techniques can efficiently classify wastes regardless of variations in the texture of the wastes that are not normally fixed in a single form in a particular environment.

Developed as part of the Machine Learning (BAXI 2223 / BITI 2223) course, this project applies image recognition and object detection techniques to bridge the gap between advanced AI technologies and practical environmental applications. The proposed waste classification system utilises automated detection to reduce reliance on manual sorting and improve consistency in waste segregation.

By incorporating object detection algorithms with an easy-to-use interface, this system enables visual feedback and real-time classification for waste sorting. Overall, the outcome of this project proves that image recognition technology has the potential for utilization in waste management.

III. OBJECTIVE

1. To automate waste sorting by accurately detecting and classifying waste items in real time using the YOLOv11 Medium object detection model.
2. To improve the accuracy and reliability of waste classification using confidence values, detection outputs, and basic statistics.
3. To provide a user-friendly and accessible web-based interface for effective waste classification

IV. PROJECT SCOPE AND SIGNIFICANCE

Project Scope:

This project focuses on developing a deep learning-based waste classification system using object detection techniques to automate waste segregation and improve recycling efficiency. The following outlines the key areas of focus :

1. **Target Audience**

   It targets households and individuals seeking appropriate practices for segregating wastes to assist recycling and minimize environmental impacts. By visually feeding back and clearly classifying wastes, the system helps users understand various classes of wastes and encourages correct disposal behavior in everyday life. The system makes waste separation easier for households and helps ensure proper disposal. Waste management organisations can also make use of it as a basic tool to improve sorting efficiency. Overall, manual work is reduced and waste management processes become more automated. In general, the system reduces manual work and supports automated waste management.

2. **Core Functionalities**

Image-Based Data Analysis (Image Input):

Users upload waste images into the system for analysis. The images are checked one by one to

find waste items using visual information. This step makes sure that the waste objects are identified clearly before classification.

Image-Based Data Analysis (Video / Live Camera Input):

Apart from static images, the system also supports video streams and live camera input. During live operation, waste items are detected continuously as the video plays. This allows the system to observe waste objects over time and maintain consistent detection results while running in real time.

Waste Classification :

After waste objects are identified, the YOLOv11 model is used to classify them into Recyclable or Organic categories. The detected items are shown with bounding boxes and confidence values. To avoid incorrect results, detections with low confidence or overlapping boxes are removed so that only clear and reliable results are displayed.

User Interface :

- A simple user interface is provided so that users can upload images or use live camera input easily.
- The classification results are displayed clearly, allowing users to understand the output without difficulty.

3. **Machine Learning Techniques**

· The YOLOv11 object detection model is used in this system for waste classification because it provides a balance between accuracy and real-time performance. Training is done using labelled images of wastes from openly available open-source datasets containing bounding boxes and class labels. The YOLOv11 model will be able to detect multiple items of wastes within the frames of one image, which will be helpful when different types of wastes are coming out simultaneously. In one step, object detection identifies the items of waste, analyzes their visual features, and classifies them into predefined categories.

4. **Platform Accessibility**

· The Streamlit application can be executed using a web browser and can be operated using laptops, tablets, and smartphones that have camera capability. It is also possible to run the application locally for testing or development but depending on user needs.

Significance:

The proposed system is able to overcome the current challenges in waste management in an efficient and effective manner. The system does so by using deep learning and object detections methods. This increases the efficiency of waste segregation and recycling. Key contributions include:

1. Improved Waste Classification Accuracy

   The YOLOv11 Medium model is used to detect and classify waste items in real time, which reduces misclassification and improves sorting accuracy. This helps reduce misclassification and improves overall sorting accuracy.

2. Enhanced Recycling Efficiency

   When waste is classified correctly, recycling efficiency can be improved through proper waste segregation.

3. Waste Management Support

   The system presents detection results and visual feedback to help users make operational decisions.

4. Cost and Resource Efficiency

   Manual waste sorting can be significantly reduced, making the processes involved in waste processing less costly in terms of workforce.

5. Transparency and Reliability

   Bounding boxes, color coding of categories, and confidence levels allow transparency and trust to be established regarding the accuracy of the classification output.

6. Foundation for Future Advancements

   The proposed system offers an avenue for future improvements such as real-time deployment, its use in conjunction with smart recycling bins, analytics, and large-scale waste management.

- Data Collection and Integration

  In this work, waste images are obtained from publicly available sources and prepared for use in training and evaluation. The dataset contains a variety of wastes including plastic waste, paper waste, glass waste, metal waste, organic waste, and general waste. Before the initiation of model training and testing activities, the images are subjected to various preprocessing techniques like image resize and normalizing. These activities are performed in order to ensure that the input variables are compatible with the YOLOv11 model.

V. PROPOSED SOLUTION

VI. RESULTS

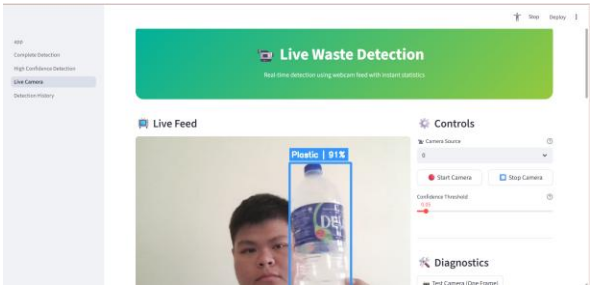


Figure 1: Real-time inference output demonstrating the capability of the system to localize and classify a plastic bottle as recyclable waste with a confidence score of 91%

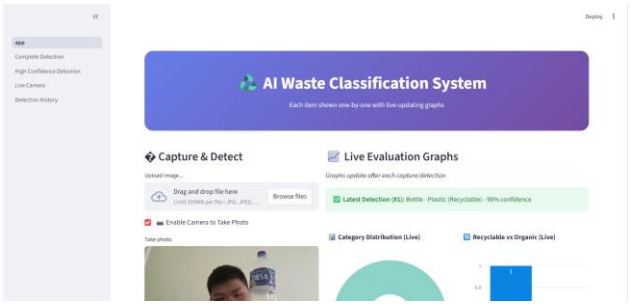


Figure 2: The integrated graphical user interface (GUI) featuring live image capture functionality, real-time probability distribution graphs, and category-specific classification metrics.

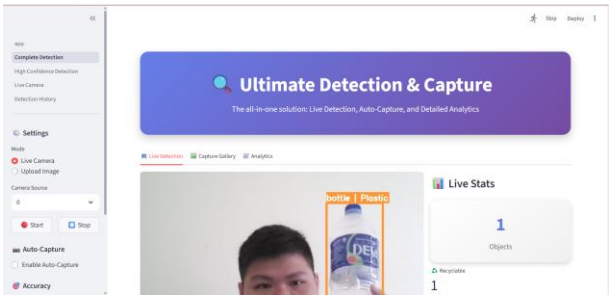


Figure 3: The Ultimate Detection & Capture dashboard is integrating real-time video inference with automated image capture controls and live object counting statistics to provide immediate quantitative feedback.

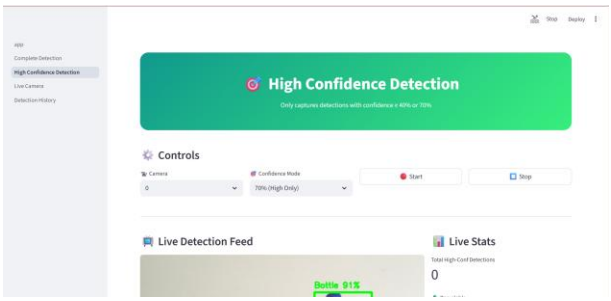


Figure 4: The high-confidence detection interface featuring adjustable threshold controls, specifically illustrating the system's filtering mechanism designed to reject predictions below 70% certainty to minimize false positive classifications.

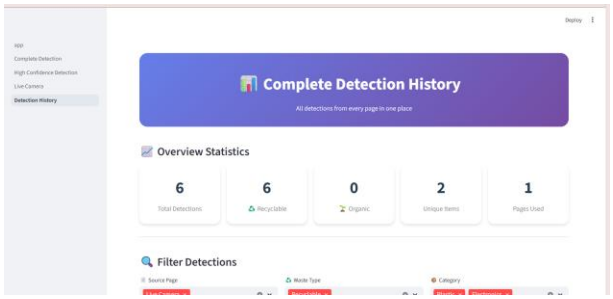
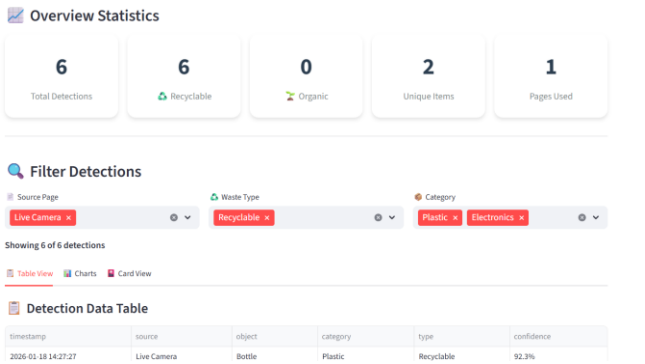


Figure 5: The historical analysis dashboard provides a centralized log of all classification events, featuring aggregate performance statistics and multi-parameter filtering controls for detailed data auditing.

Figure 6: The detailed analytics interface displaying the results table that lists every single item the system found that allows it to check the exact time, waste category, and accuracy score for each one.



VIII. CONCLUSION

The proposed work designed a real-time waste classification system using the YOLOv11 algorithm for object detection. The system helps in efficient waste management using object detection algorithms. The system has the capability of scanning images and classifying the waste in real time. Performance analysis of the system was conducted using evaluation metrics. Furthermore, the system was designed such that the results of object detection could be displayed using web technology.

In order to further optimize this system, some IoT-based sensing devices like weight sensors or infrared sensors can also be included. This will make it possible to automatically scan the trash and collect data from intelligent recycling bins in real-time. Thus, this system will be able to perform its functions as a complete IoT-based trash management system. Apart from this, additional improvements may be done by including extra categories for trash and using data analysis on a larger scale in support of trash management in the city. The detection model need not be updated in the same way. The latest versions of this model, such as YOLOv13, can be implemented in the future.

Video Youtube link : https://youtu.be/In-1HWGYCNA

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